

2023



Improving Performance by 30% for 400 Million Daily Transactions Using Amazon EBS with McAfee

Learn how McAfee in the cybersecurity industry reduced backup time by 50 percent using Amazon EBS io2 Block Express volumes.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

30%

performance improvement for 400 million daily transactions

81%

reduction in reporting time

99.95%

availability achieved across McAfee ser

Unlocked capabilities

Hi, I can connect you with an AWS representative or answer questions you have on AWS.

Powered microservices

Need more info? Highlight any text to get an explanation generated with AWS generative AI.



Online protection company [McAfee](#) was growing year over year, but hardware issues at its on-premises data centers prevented the company from meeting ambitious targets for availability at scale. McAfee wanted to deliver the robust reliability needed to serve hundreds of millions of customers globally, so it committed to going all in on the cloud. "On our legacy infrastructure, we delivered between 99.44 and 99.94 percent availability, but our target was 99.95 percent availability across our services," says Srinivas Sigatapu, senior director for the consumer DevOps and site reliability engineering team at McAfee.

After considering several cloud providers, McAfee decided to embark on a lift-and-shift migration to Amazon Web Services (AWS). The company planned carefully to verify that its customers would not experience any disruption during the migration of over 2 PB of data. Then, starting in late 2021, it migrated its critical SQL workload to [Amazon Elastic Block Store](#) (Amazon EBS), which provides easy-to-use, high-performance block storage at virtually any scale. "We achieved our target of 99.95 percent availability for 500 million devices by migrating to AWS," says Srinivas.



Opportunity | Using Amazon EBS io2 Block Express to Achieve Optimal Performance and Speed in the Cloud

McAfee was founded in 1987, and the company is focused on protecting people, not just devices, empowering them to live securely through integrated, intuitive solutions. McAfee secures its customers' digital lives by protecting identities, providing virtual private network connections, and enhancing privacy. When McAfee saw rapid growth in 2020, the company had already been exploring cloud services as it planned for future expansion. It began using AWS services for some of its new products.

After a positive initial experience, McAfee decided to migrate its business-critical [SQL database](#) to AWS. “We chose AWS because our team already had the necessary skill set, the costs were lower than other offerings, and AWS has multiple [AWS Regions](#) for our operations,” says Srinivas. After several months of planning, McAfee migrated the database to [Amazon EBS io2 Block Express volumes](#), which offer block storage in the cloud with four times higher throughput, input/output operations per second, and capacity than io1 volumes, along with sub-millisecond latency. McAfee is seeing significantly better speed and performance since the migration. “Using our on-premises infrastructure, reporting took 4 hours, but we reduced that to as little as 45 minutes by migrating to AWS,” says Lohith KS Chalam, lead database administrator at McAfee.

Solution | Driving a 30 Percent Performance Improvement for 400 Million Daily Transactions Using Amazon EC2 and Amazon EBS

Throughout its migration, McAfee worked closely with AWS to verify that it had the right plan in place. The company engaged in multiple reviews using [AWS Well-Architected](#), which cloud architects can use to build secure, high-performing, resilient, and efficient infrastructure for a variety of applications and workloads. It also bounced ideas off of AWS support professionals to find the optimal services for its use case. McAfee used [Amazon Elastic Compute Cloud](#) (Amazon EC2), which offers secure and resizable compute capacity for virtually any workload, to power its critical workloads in the cloud. More specifically, to optimize performance, it used the [AWS Nitro System](#), a combination of dedicated hardware and lightweight hypervisor for faster innovation and enhanced security. “We shared our requirements with AWS for our critical workload and data warehousing workload, and AWS recommended Nitro-based Amazon EC2 instances, which have met our needs,” says Saravanakumar Karunakaran, senior architect at McAfee.

Using Nitro-powered [Amazon EC2 R5b Instances](#), which are the next generation of memory-optimized instances for Amazon EC2, as well as [Amazon EC2 X2i Instances](#), which offer up to 50 percent better performance than X1 instances for memory-intensive workloads, McAfee boosted performance by 30 percent across 400 million daily transactions. “After migrating, our applications were flying along, and we haven’t seen any issues with storage or computing,” says Saravanakumar. Although McAfee enhanced performance across many dimensions, improvements in querying have been particularly impactful for both internal teams and customers. Login times have reduced significantly, leading to increased customer satisfaction. Also, overall costs have fallen by about 20 percent, grounding a more seamless experience across McAfee’s global customer base.



Now that its workloads are running on AWS, McAfee teams can more easily carry out enhancements without disrupting service for customers. “With our legacy infrastructure, failovers used to take 5 minutes,” says Srinivas. “Now, we’re consistently seeing less than 90 seconds of failover time.” Moreover, backup time for critical workloads has fallen by 50 percent using Amazon EBS io2 Block Express volumes. For noncritical workloads, McAfee uses [Amazon EBS General Purpose Volumes](#), which are solid state drive volumes that offer balanced cost and performance for a majority of workloads, to deliver greater performance and reduce costs.

McAfee achieved significant overall cost savings by migrating to AWS. “When we look at hardware costs and add the operational costs of the teams involved in running an on-premises data center, we have saved significantly by running our systems in the AWS Cloud,” says Srinivas.

Outcome | Reducing Maintenance Work by Using Amazon RDS

Moving forward, McAfee plans to keep reducing the workload on its teams and will migrate many important workloads to [Amazon Relational Database Service](#) (Amazon RDS), a collection of managed services that makes it simple to set up, operate, and scale databases in the cloud. “We meet with AWS almost biweekly to review new releases, improve our services, and reduce our operational overhead,” says Srinivas.

Now that its critical SQL workload is on AWS, McAfee is focused on advancing the customer experience. “With the money we’ve saved by using AWS, we are investing in new products and services for our customers,” says Srinivas.

About McAfee

McAfee is a leading online protection company that strives to make life online safe and enjoyable for everyone. McAfee delivers comprehensive protection to safeguard people’s privacy and identity in addition to its award-winning antivirus solution.

AWS Services Used

Amazon EBS io2 Block Express volumes

Amazon Elastic Block Store (Amazon EBS) is an easy-to-use, scalable, high-performance block-storage service designed for Amazon Elastic Compute Cloud (Amazon EC2). EBS io2 Block Express volumes are designed to deliver per volume performance of up to 4,000 MB/s throughput, 256,000 IOPS, 64 TiB storage capacity, and 1,000 IOPS/GB, as well as 99.999% durability and sub-millisecond latency.

[Learn more »](#)



Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 700 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

AWS Nitro System

The AWS Nitro System is the underlying platform for our next generation of EC2 instances that enables AWS to innovate faster, further reduce cost for our customers, and deliver added benefits like increased security and new instance types.

[Learn more »](#)

AWS Well-Architected

AWS Well-Architected helps cloud architects build secure, high-performing, resilient, and efficient infrastructure for a variety of applications and workloads.

[Learn more »](#)

More Software and Internet Customer Stories

GERMANY



HeyJobs on AWS

ITALY



Namirial Streamlines
Contract Signing with...



HeyJobs aims to make job searching and recruitment more effective for candidates and employers.

2024

Namirial enhanced its eSignAnywhere solution with generative AI using



2024

COSTA RICA



Modernizing Applications in Minutes Using Amazon Q...

Learn how Novacomp accelerated



2024

AUSTRALIA



Dovetail Enhances Customer Insights with...

Learn how Dovetail accelerates development of new generative AI products and features using Amazon Bedrock, deploying the latest large language models while reducing security and administrative overhead.

2024

1 2 3 4 5 6 ... 201

Get Started

Organizations of all sizes across all industries are transforming their businesses and delivering on their missions every day using AWS. Contact our experts and start your own AWS journey today.

Contact Sales

Resources for AWS

Getting Started
Training and Certification
AWS Solutions Library
Architecture Center

Developers on AWS

Developer Center
SDKs & Tools
.NET on AWS
Python on AWS

Help

Contact Us



2

Ticket

AWS re:Post

Learn About AWS

What Is AWS?

- What Is Cloud Computing?

AWS Accessibility

AWS Inclusion, Diversity & Equity

What Is DevOps?

What Is a Container?

What Is a Data Lake?

What is Artificial Intelligence (AI)?

What is Generative AI?

What is Machine Learning (ML)?

AWS Cloud Security

What's New

Blogs

Press Releases
- Product and Technical FAQs

Analyst Reports

AWS Partners
- Java on AWS

PHP on AWS

JavaScript on AWS
- Knowledge Center

AWS Support Overview

Legal

AWS Careers



Amazon is an Equal Opportunity Employer: *Minority / Women / Disability / Veteran / Gender Identity / Sexual Orientation / Age.*

Language

- | عربي
- | Bahasa Indonesia
- | Deutsch
- | English
- | Español
- | Français
- | Italiano
- | Português
- | Tiếng Việt
- | Türkçe
- | Русский
- | ไทย
- | 日本語
- | 한국어
- | 中文 (简体)
- | 中文 (繁體)

Privacy

- |
- Accessibility
- |



